

Intelligent Deep Learning Systems for Predicting Customer Purchase Intentions in Phygital Retail Industry

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Abstract -- The concept of phygital retail promotes the integration of digital and physical channels toward enhancing customer engagement, hence satisfaction. This research tried to know whether the intelligent deep learning system would be able to predict customer purchasing behavior in a phygital environment. In this paper, high data that is comprised of touchpoints-both digital and physical-were implemented into various deep learning models which included ANNs, CNNs, RNNs, and LSTM. In this research, it has been depicted that both RNN and ANNs become the best model due to the highest accuracy as well as maximum AUC values which range at 92.70% and 0.97, respectively. Another insight that has come out of this visualization study is how correlation matrices and distribution plots highlighted the need to marry digital copies of online metrics like time spent or visits with the ideal customer journey. The models, through advanced data preprocessing and feature engineering techniques, were highly predictive. This study fills the gap in existing literature by highlighting the integration of deep learning into phygital retailing offering actionable strategies for personalized marketing

Keywords: *Deep learning, Prediction, Phygital, Purchase Intention*

I. INTRODUCTION

Phygital retail is the seamless integration of both physical and digital experiences into marketing to better enhance customer engagement and satisfaction. The strategy draws strength from the two environments of in-store and online, creating a holistic and engaging experience for the shopping journey. For instance, in a physical location, a customer can interact with a product through digital kiosks in the store or an augmented reality application, thus experiencing something in a virtual sense. It must be all instantaneous, connected, and engaging, all part of a model that is fast-evolving according to the expectation of a modern consumer. Deep learning is the constituent of the machine learning section of AI. The technique used in data modelling is a multilayered network to pick up complicated patterns. Deep neural nets are "inspired" by how a human brain is "designed" to work, so computers can recognize images, speech, and even make decisions for themselves while processing natural language. Unlike traditional machine learning models, where feature extraction typically needs to be done manually, deep learning models automatically discover

features from raw data through multiple layers of abstraction, which allows them to reach very high accuracy on tasks such as image classification. Lower layers could detect edges and higher layers can identify complex structures like faces or objects. Because deep learning has gained in effectiveness with the development of large datasets and improvements in the computational power, especially after the appearance of graphics processing units, it is now possible to train deep networks that have multiple layers. Through these developments, several applications of AI have had breakthroughs.

As deep learning is one part of machine learning, it's actually slowly making a shift for retailers to become better and more highly advanced when analysis and automation go hand in hand with data. Big data findings about retail today can be further applied to an algorithm by taking advantage of the intelligent systems with various applications as providing personal recommendations to checking the stocks or any sort of optimization pertaining to clients' satisfaction. For example, setting the right kind of products for client preference as analyzed in any given purchase related detail. This therefore becomes useful to sell and bring improvements in satisfaction. It also helps in dynamic pricing, where algorithms change prices in real-time based on demand, market conditions, and customer behaviour to maximize revenue. In the case of inventory management, deep learning predicts demand, and it allows retailers to optimize stock levels and avoid overstocking or understocking. Retailers also use chatbots and virtual assistants powered by deep learning to offer all time customer support and shopping assistance through natural language processing. In addition, deep learning is used in fraud detection systems to detect unusual patterns in customers' transactions, thus preventing fraud. This goes even deeper into the use of deep learning, which includes demand forecasting. Therefore, the comparison of the production and supply chain with customers' demands means things are getting increasingly efficient. As a matter of fact, companies like Amazon have been using generative AI in their offerings to make the shopping experience richer. This revolutionizing technology allows the retailers to get into personalized customer experience while driving the efficiency of operation in this new world of data-driven retail. Deep learning uses predictive analytics, natural language processing, and dynamic optimization for retailers to play a

winning role in a very dynamic market, thus meeting emerging needs.

Deep learning indeed has revolutionized the phygital retail world with the world of physical and digital shopping being effortlessly brought together so as to better customer engagement as well as operation efficiency. Using deep learning algorithms in analysing big data generated through both in-store and online activity, it lets retailers deliver their services on the personal level with optimized inventory management. For instance, computer vision and deep learning methods allow a customer to browse the catalogues interactively and thus bridge the gap between the world and digital reality. Deep learning also offers support for simulating the in-store journey of the customer. This can further also be used to optimize store layouts and fully personalized marketing practices. Therefore, these technologies can allow the phygital retail environment to offer an integrated and immersive experience for shoppers. It satisfies the changing expectations of modern consumers. With deep learning incorporation in the phygital retail, customer satisfaction with the business increases along with the efficiency with which businesses operate, thereby making the retailers compete favorably within a competitive landscape.

This paper attempts to understand the concept of the integration of intelligent deep learning systems for predicting the purchase intentions of customers in a kind of phygital retail environment that integrates physical and digital experiences. This is achieved by considering the advanced application of algorithms in ANNs, CNNs, RNNs, and LSTM models in the interpretation of customer behavioural patterns for making shopping more personal. Overall, the purpose of this research is to create predictive models that can help retailers enhance decision-making processes while increasing customer satisfaction and improving their operations. In addition, the paper goes on to talk about how data-driven insights link the slit between the physical place and digital space for seamless spending. Doing so, we also hope to join the growing volume of literature about applying deep learning to phygital retail for strategic advantage.

II. LITERATURE REVIEW

The deep learning techniques indicated outstanding potential for the retail sales forecasting and purchase intentions prediction. They have enhanced the stock management and marketing strategies. According to (Razin, 2024) [10] hybrid model CNN- LSTM outperformed the stand-alone models CNN and LSTM as the hybrid model indicated the lowest values of RMSE and MAE values for the given model for sales forecasting. That brings the hybrid approach closer to replicating spatial as well as time-series patterns in better ways. Comparing linear models against deep learning technique like RNN and LSTM as applied in a sales forecasting process for revenue optimization is done. (Shriram, 2023) [11]. This research was founded on the basis that deep learning is an upgrade of managing inventory better as it relates to knowing customer behaviour and trends within the market. This work was centered on the fact that deep learning is an enhancement to the management of inventories through better knowledge of customer behavior and market trends. (Zhihong, 2024) [15] has elaborated this by showing how RNNs catches up on the dependencies in e-commerce user actions that can make predictions considerably better than some like logistic regression and decision trees. This modelling technique

captures the rich and complex user behaviour that is mainly in enhancing prediction accuracy within an e-commerce setting.

(Xiaoye, 2023) [12] presented a BERT-LST Net-Soft max model designed to determine the sentiments of the consumer toward a particular product as well as establish purchasing intent. Indeed, it had excellent work with the tremendous usage of being used as a better tool in helping describe the consumption pattern and aid retailer decisions. (Mamadov, 2023) [6] applied models of deep learning for the prediction of online customer purchasing intent. This attained an accuracy of 90.24% and a validation score of 88.15%. It brought such real-time application of models into offering personalized offers to non-purchasers and rewards for loyal customers. (Neha, 2021) [7] showed importance in platform engagement and characteristics of customers. The author further illustrated that deep learning surpasses the traditional approaches of machine learning for realistic purchase prediction in the retail scenario. This innovation prepares the developers of platforms for more engagement (Yifan, 2024) [13] introduced the Deep Adaptive Evolutionary Ensemble (DAEE) which is a grouping model of evolutionary ensemble learning and deep forest algorithms introduced for the first time. This resulted in an increase in predictive accuracy, robustness, and training speed, in itself six times faster than baseline models with an increase of 5.02% AUC value. (Eshrak, 2022) [3] conducted a deep literature review and explained how CNN and LSTM can make high accuracy with a prediction of purchase intention up to 98% by handling large datasets with complex features, far superior than the traditional machine learning approach in managing consumer relationships. Recently (Bagus, 2023) [1] developed Fast Text on the case for selling property amidst the pandemic ending with a 94.19% accuracy result and 92.03% AUC prediction of consumers buying intentions. The same research further attached the model in a web application so real-world applicability will be expected from it. The study further included the model in a web application, so real-world applicability can be envisioned. (Ray- I, 2022) [9] recently used LSTM networks in an application of prediction of purchase intents from ECG data created by smartwatches. After data augmentation, the accuracy was reported as 82.1%. This innovative approach explains how deep learning can be flexible in consumer behavior understanding. (Yiran, 2024) [14] applied CNNs to the analysis of brain activity signals from ERP experiments that resulted in 98% accuracy in the selection of consumers. This study showed how deep learning can be used in the production of consumer insights through the analysis of physiological signals where CNNs can catch presentation order effects. (Chao, 2024) [2] applied the CNN and LSTM models on data from an e-commerce website, images, and text for the prediction of purchasing preferences and consumer emotions for the improvement of personalized recommendation. (Humphrey, 2018) [4] presented a model of an RNN to capture event dependency across user sessions without heavy feature engineering, achieving a classification accuracy of 98%. The study enhanced its performance by reversing the order of user sessions to demonstrate how flexible RNNs are to work with varying session lengths. (Khang, 2018) [5] proposed a Wide and Deep Learning Network using mouse click events to predict customer behavior on purchases. This model was able to get an accuracy of 78.26%. It had outperformed other deep learning methods using low end and high-end interactions of key features. These studies represent the boundary of how far deep learning can change the face of retailing and e-commerce: from purchase intent prediction to optimizing inventory and fine-tuning personal consumer experience. The author (Ranjith, 2024) [8] developed a predicting model based on the Long Short-Term Memory network to forecast churn in the telecom industry by making use of 4,250 customer profiles that are used in finding sequential data on the pattern of churn, and attains an accuracy level

of 95.3%. These findings would improve the retention of customers and illustrate how the LSTM networks could be applied for solving churn prediction problems in turn enhancing the strategic decisions managing the customer relationship and also retaining the existing customers.

Though a tremendous amount of success has been obtained in deep learning applications within retail, surprisingly minimal research is applied to how these can be incorporated into phygital retail spaces, which encompass both the tangible and digital realities of the experience. All work done previously pertains only to either offline or online, omitting all these complexities that the integration would encompass. Deep learning models appear promising towards customer purchase intention predictions; yet, there still lacks holistic frameworks for dynamic, interactive phygital spaces. The integration of real-time channels in-store as well as those digital remains rather under-explored and has led to the knowledge of lesser comprehensiveness about the customer from such channels. This study fills such gaps by designing intelligent deep learning systems especially for the phygital retail context with the intention of optimizing customer engagement and business outcomes.

III. METHODOLOGY

A. Dataset Structure and Key Attributes

The dataset used for this study provides a good basis to examine the customers' behaviour in a phygital retailing environment and predict the purchase intentions. It includes the entire range of variables, covering both physical and digital touchpoints, so the opportunity is given to examine the way they interlink in the processes of purchase intention. The key variables are abandoned cart and product returns, a clear indicator of the customers' dissatisfaction or disengagement, and the time spent by customers online and offline captures their fluid integration with digital and physical experiences. Other quantifiable measurements in terms of total cart value, frequency of visit, and total purchases all add up to their engagement in purchases, but the recency of purchase will depict an engagement pattern. The dataset also has demographic attributes such as age, gender and income level which aids in segmentation. Behavioural indicators like CLV and response to discounts along with sentiment related features like positive reviews and negative reviews are also included in this dataset. The target variable purchase intention is the intent of a client to buy at a given period. This well-suited set of data perfectly models prediction using classifying algorithms since the target is binary. It covers a wide range of variables including both customer behaviour and demographics and hence can give a panoramic view of the customer journey in a hybrid retail landscape.

B. Dataset Preprocessing

To ensure the readiness of the data for analysis and modelling some preprocessing steps were implemented. These steps play crucial role in enhancing the quality of data, minimize biases and maximize model performance. The first step is to handle the missing values with the help of the `isnull()`. In the missing places numerical variables were assigned with their mean or median values. Data was standardized using the library `StandardScaler()` and `Normalize()` were used to min-max scaling. With the help of Interquartile Range outliers

were detected and treated. Dataset was split into two training (80) and testing (20) to prevent overfitting.

C. Tool and Model

Jupyter Notebook is an open-source web application that facilitates developing and sharing live documents that may contain code, equations, visualizations, and even narrative text. More than 40 programming languages, such as Python, R, and Julia, are used for support towards a broad array of data analysis, scientific research, and deep learning projects. This environment, therefore, invites direct coupling between code execution and documentation to boost reproducibility and collaboration within computational workflows. Artificial Neural Networks, or ANNs, are the imitation of neural networks and work of the human brain, formed by layers of interconnected nodes or neurons, for the purpose of recognizing patterns or solving complex problems by the application of models on input data. ANNs represent the base for deep learning, and they find extensive applications in image and speech recognition. CNNs are specific types of ANNs designed to process data that has a grid-like topology. It could be an image. There are convolutional layers for automatically learning adaptive hierarchies of features spatially and making them highly effective in recognizing images and videos. Among the very effective subclasses for modelling sequential data, recurrent neural networks capture the information from earlier steps through their memory states; they are excellent tools for spotting patterns that exist diffused across time, therefore suitable for either time-series predictions or language model applications. Such an improved version of an RNN would be called an LSTM network. To solve such limitations of long-term dependencies while capturing the hidden representations over several extended sequences within models and methods.

IV. FINDINGS

A. Exploratory Data Visualization

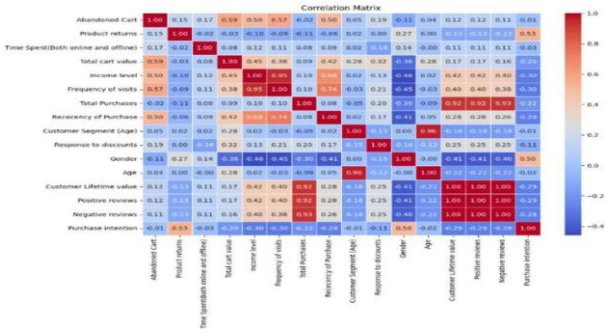


Fig 1. Correlation Matrix of the Key feature

Correlation matrix highlights the relationship between various factors that are influencing customer purchase intention in a Phygital retail environment. The target variable Purchase Intention reveals positive correlations with positive reviews, total cart value, frequency of visits and total purchases indicating the significance of constant engagement across various digital and physical touchpoints. On the other hand, abandoned carts and negative reviews exhibit negative correlations with purchase intention indicates that improving both online and in-store customer experience can mitigate these deviations. Personalized loyalty programs and targeted promotions can enhance purchase behaviour as the customer lifetime value and response to discounts show moderate correlations.

On the whole, the correlation matrix underscores the interplay between physical behaviour such as frequency visits and digital metrics such as time spent online and reviews in shaping customer shopping intention in hybrid retail landscape.

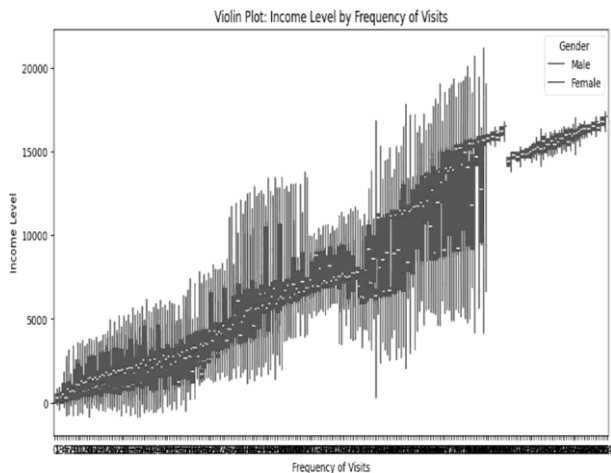


Fig 2. Violin Plot of Income level and Frequency of visits

Violin plot demonstrates the relationship between the frequency of visits and income level to a retail environment which is segmented by gender (male and female). Higher income group correspond to an increase in the frequency of visits for both male and female highlighting positive correlation. The results suggests that individuals with higher income engage more frequently with both physical and digital channels and the reason behind this is higher purchasing power and quest for various shopping experiences. In the Phygital retail context signifies the need for tailoring engagement strategies according to the income of consumers. Higher end group people who visit frequently may benefit form premium in store services, exclusive offers and discounts and a tailored online recommendation to enhance their experience. On the other hand, strategies are to be developed to make them visit frequently by providing discounts and rewards for their purchase which might help in bridging engagement gaps.

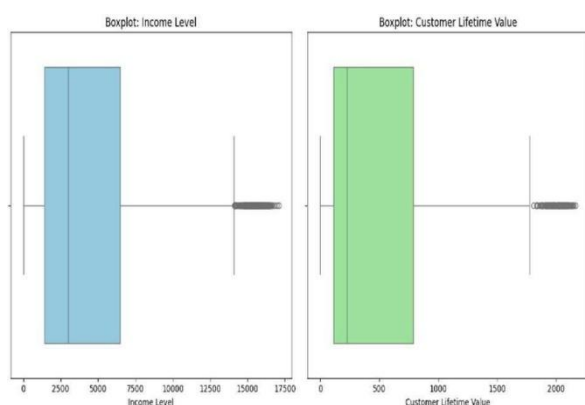


Fig 3. Pair Plot of Time spent and Total Purchases

The pair plot reveals the connection between the time spent both in online space and offline place with total sales in the context of purchase intention within a hybrid retail environment. Customers with higher purchase intention are spending less time when compared to the customers without purchase intention. This suggests that conversions are driven

by efficient and seamless shopping experience. Large number of customers fall into the category of moderate time-spent and minimum purchase category highlighting the possible inefficiencies or deviations in the process of purchasing. As the time spent increases, return on purchase decreases revealing the significance of optimizing the online and offline touchpoints in speeding the decision-making process. Results establishes that it is important for the physical retailers to employ AI-driven analytics and personalization in enhancing customer satisfaction.

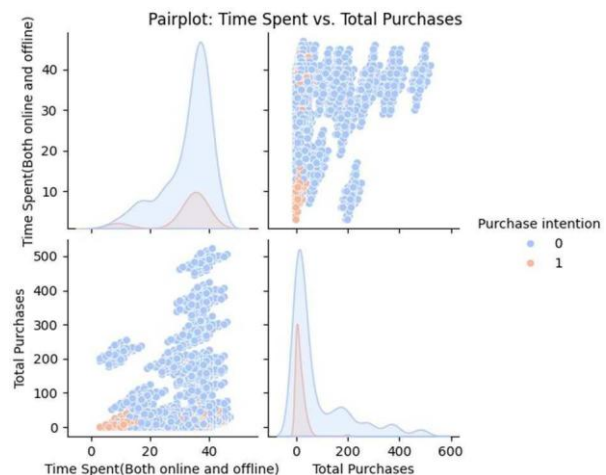


Fig 4. Boxplot of Income level and Customer Lifetime Value

Boxplot of Customer Lifetime Value and Income level reveals the key understandings for phygital retail strategy. From the plot we can identify that majority of the customers falls under the mid-range of income level along with a few high-income levels. To cater various segments of customers, the retailers can adopt distinguished marketing strategies which makes the actual customers to become a loyal customer. The above plot also shows that maximum customers belong to the medium range of customer lifetime value and only very few high value customers are encountered. These high value customers are to be taken care with providing mutual benefitted loyalty programs as well as personalized experiences.

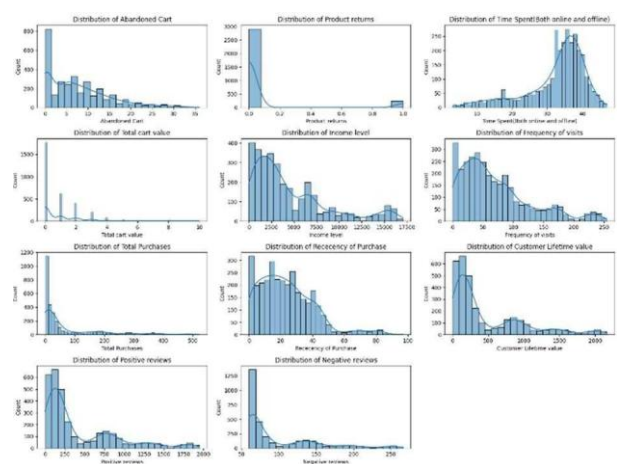


Fig 5. Histograms of the key features

The visualizations above offer distributions of key features

within the dataset. Customer behaviours and attributes in the phygital retail environment are also highlighted here. Features "Abandoned Cart," "Product Returns," and "Total Purchases" seem to be significantly skewed, so most customers' engagement is actually low in those areas. The two other variables in this case were "Income Level" and "Time Spent," which are also right-skewed. This means that most customers have specific income groups and like to spend less time shopping, but the outliers are those who spend a longer period of engagement. The distributions of "Frequency of Visits" and "Recency of Purchase" show that the retention of customers and shopping intervals vary, which may impact phygital strategies such as personalized engagement or targeted promotions. Distributions of "Customer Lifetime Value" and "Positive Reviews" point out that customers are generally small contributors, whereas high-value, highly satisfied customers are the only touchpoints strategic focus should be on.

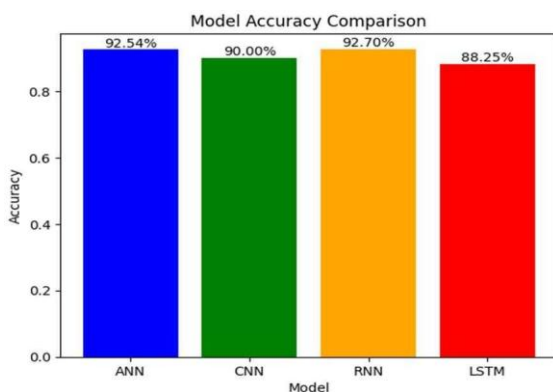


Fig 6. Comparison of Accuracy Scores for the models

The above bar chart is the comparison of four deep learning models: Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Recurrent Neural Network (RNN) and Long Short -Term Memory (LSTM). The Recurrent Neural Networks achieves the highest accuracy at 92.70% followed by the Artificial Neural Networks at 92.54%. Comparatively the accuracy scores are low for Convolutional Neural Network and Long-Short Term Memory at 90.00% and 88.25% respectively. This analysis reveals that Recurrent Neural Network and Artificial Neural Network are the best fit models for predicting the purchase intention of the customers outstripping Convolutional Neural Networks and Long Short-Term Memory.

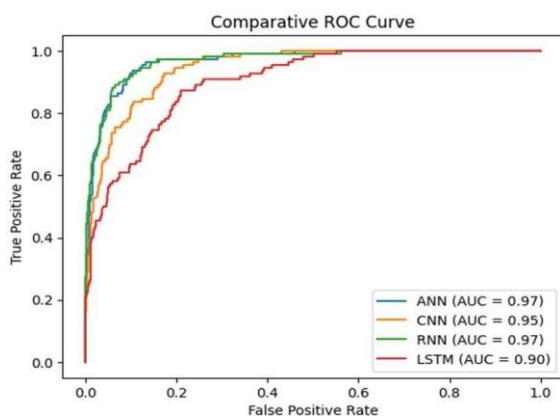


Fig 7. Receiver Operating Characteristics Curve

Customer Purchase Intention within a Phygital retail Environment can be determined with the comparative ROC curve which highlights the predicting performance of various deep learning models like Artificial Neural Networks, Convolutional Neural Networks, Recurrent neural Networks and Long Short-Term Memory. Comparison of all the four deep learning model reveals that Artificial Neural Networks and Recurrent Neural networks has achieved the highest accuracy score in prediction as well as highest Area Under Curve (AUC) of 0.97. This result specifies their sturdiness in capturing complex patterns from combination of both Digital and Physical customer interactions. Convolutional Neural Network's AUC is slightly lower with 0.95 indicating their wellness in processing spatial or image-based data that includes augmented reality interfaces and product images. Long Short-Term Memory performed comparatively lower with an AUC of 0.90 signifying the shorter customer interaction data sequences can also be common in this context. The results highlight the appropriateness of Artificial Neural Networks and Recurrent Neural Networks for predictive tasks in Phygital retail, where customers are influenced by an interplay of both digital and physical experiences.

V. CONCLUSION

This research study, therefore, outlines the transformative nature of intelligent deep learning systems with regard to forecasting customer purchase intent in phygital retailing. The best models, being Recurrent Neural Networks (RNNs) and Artificial Neural Networks (ANNs), where the integration between physical and digital touchpoints resulted in superior effectiveness and accuracy performance. It signifies that the convergence of digital measures such as in-store frequency is indeed critically important and thus needs harmonization with those more traditional and observable, such as physical indicators of engagement. In summary, using the findings provided can help retailers give more targeted experience, more effective operations, and data-driven approaches to decisions. Such study bridges long-known gaps within extant literature, providing a holistic framework that brings deep learning towards hybrid scenarios for retailing applications. Such results not only align with the emerging expectations of customers but also provide the business with competitive edge in transforming times. On the bottom line, this study sets the wheels in motion towards creative AI-inspired approaches that redesign the engagement experience of customers for new retail today.

VI. FUTURE SCOPE

Some potential future work involves the development of hybrid deep learning models that would integrate CNNs and RNNs to boost the accuracy of predictions. Additionally, the real-time data streams from wearable devices and IoT sensors will be of great help in enriching knowledge about customer behaviour. The research can be generalized by expanding the study to other retail domains and global markets. Other directions of future research could be on the issues of data privacy and fairness in AI applications. These implications demonstrate the vital role of technology in reshaping the retail landscape.

VII. PRACTICAL IMPLICATION

The results obtained in this paper are very essential for phygital retail and the design of AI-driven insights with regard to personally designed loyalty schemes and store layouts to heighten customer

experience. Deep learning model-based demand prediction enables precise, minimal over/understocking. Predictive analytics helps in a number of areas related to the strategic decision for competitive advantage- dynamic pricing as well as relevant promotion. Policymakers and organizations can take these findings to heart in reconciling retail strategies with sustainability efforts, in meeting consumer expectations for ethical AI usage. What these consequences bring about is how technology interferes in the shifting tides of retail

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